UNIVERSITY OF CALIFORNIA

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Theory, Design, and Implementation of Hunnid Channel EMG

A dissertation submitted in partial satisfaction

of the requirements for the degree

Doctor of Philosophy in Mechanical & Aerospace Engineering

by

Samantha Herman

Month Year

ABSTRACT OF THE DISSERTATION

Theory, Design, and Implementation of
Hunnid Channel EMG

by

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(Abstract omitted for brevity)

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Dedication yet to be decided

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VITA

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PUBLICATIONS

add publications

CHAPTER 1

Introduction

Introduction Sample Text

CHAPTER 2

Circuit Theory

Before designing any circuit, it is necessary to identify goals, restrictions, and assumptions. Since this circuit is primarily designed with the intention of acquiring bioelectronic signals (eg: electromyography) from the inside of the human body, then the overarching goal must be to develop a framework for noise-robust low-power sensing with as many channels as possible. This lends itself to a few constraints and a handful of simplifying assumptions.

First, to enable as many channels as possible, we'll have to engage with some form of multiplexing. Time-domain multiplexing, while relatively simple, needs to scale readout speed as the number of channels increases, and there are challenges with the readout. To date, approaches in this direction have involved time domain multiplexing by means of inductive coupling of an externally mounted coil with discrete internal coils [TODO: cite Weir et al], or the implantation of a base station with infrared laser telemetry [TODO: cite mira, also double check how they do telemetry] both of which suffer from issues involving scalability, sensor size, and generalizability. By comparison, frequency-domain multiplexing seems the more attractive option, especially considering that capacitive coupling through the skin has a bandwith of acceptable wireless power transfer between 50-200 MHz [TODO: cite capacitive coupling paper]. We estimate that frequency-domain multiplexing would enable an increase in channel count as compared to time-domain multiplexing techniques by at least an order of magnitude [TODO: once you get to nanofab section of work, come back with a more accurate estimate].

Second, bioelectronic signals need to be measured differentially. This is because the ambient or background voltages (ie: local field potentials) are constantly changing due to the aggregate electrical activity surrounding the measurement area. There are two approaches which can achieve this—developing bipolar sensors, or mimicking bipolar sensors by subtracting the measurements of two monopolar sensors for every signal of interest. For several reasons, inherently bipolar sensors are the better choice—they consume half as much bandwidth, take up half as much space, involve simpler data processing to readout, and have lower requirements on sensor range. However, because some applications beyond the scope of bioelectronic signals may not require differential measurement, I will briefly outline the design of monopolar sensors in the subsequent sections.

Third, different parts of the human body (heart, brain, and muscles, to name a few) produce bioelectronic signals of different amplitudes. This means that ideally, we design the circuit so that it is easily modifiable (eg: through the change in value of one or two components) to be sensitive to a wide range of signals. For simplicity, and for the sake of minimizing power consumption during operation (or at least, for the parts of it that would operate inside the human-body to be low-power), we will try to avoid designing circuits that require a built-in bias voltage in order to function. If necessary for built-in bias voltage (mostly for the purpose of biasing transistors in gyrator-C configurations to make active inductors that fit in a smaller footprint) we limit ourselves to voltage regulators operating off the input from the transmission line.

Finally, we'll perform analysis of the circuit under the (reasonable) assumption that the bioelectronic signals we expect to measure vary on a timescale much much slower than the time constant of the circuit. Though not explicitly required by our previous goals, we'll also assume that our circuit is linear (or if it contains non-linear elements, that we'll operate in a linear regime) for reasons that will become clear in subsequent sections—namely to enable the operation of a readout scheme which requires linear circuit responses. So long as we do not violate these two assumptions, we can perform our analysis of the circuit as a steady state impedance network in the Laplace domain.

2.1 LC Circuits

With that in mind, we settle on the simple LC resonant circuit as our sensor. The principle behind their use is that LC circuits have different impedance properties near and away from resonance (namely, they can be made to have zero impedance on resonance, and close to infinite impedance at regions not too far from resonance) they are suitable for frequency division multiplexing. In other words, this property enables selective interaction—that is, sending a sinusoid near the resonant frequency of one resonator will not have any effect on other resonators with sufficiently different resonant frequencies. Identifying what distance from the resonant frequency constitutes 'sufficient' is the goal of analysis later in this chapter and is equivalent to the frequency bandwidth that a resonator consumes.

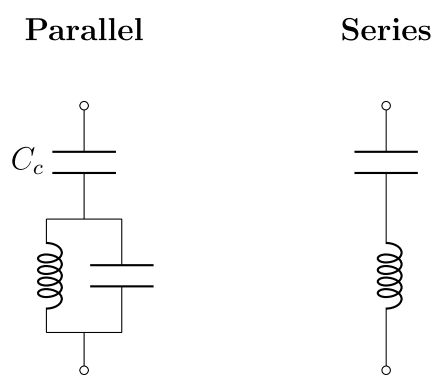


Figure 2.1: The series LC circuit has zero impedance at resonant frequency. However, the parallel LC circuit (in the absence of the coupling capacitor C_c) would have infinite impedance at resonant frequency. The addition of the coupling capacitor enables the circuit to also have zero impedance at it's resonance frequency. The value of this coupling capacitor can also play a role in tuning the resonator's bandwidth.

In this manner, a large number of LC circuits can be coupled onto a single transmission line, each excited by a unique sinusoid at each resonant frequency. In order to use the resonators as sensors, note that the resonant frequency of the circuit depends directly on the value of the inductance and capacitance in the circuit. Thus, we can use the capacitive or inductive elements in the circuit as sensors if we monitor the resonant frequency of the circuit (ie: by observing shifts in the amplitude and phase of a constant frequency sinusoid at the circuit's original resonance frequency). Finally, it is also a linear circuit, which means that a readout scheme which analyzes the change in phase and amplitude of an input sinusoid across the resonator is feasible.

[TODO: comment on ω_0 occuring when $\mathrm{Im}[Z] = 0$, see bmazin thesis sentence following eqn 2.37, also might be worth deriving how an LC circuit works or showing the eqn for resonant frequency? idk how much background is necessary]

2.1.1 Varactor Diodes

In previous sections, the discussion has been focused on generalities. Shifting gears, we will now focus on taking the established general circuit framework and using it to tackle the specific problem of acquiring bioelectronic signals inside the human body using frequency domain multiplexed LC resonators. First, it is necessary to determine whether or not the inductance or the capacitance of the LC circuit ought to be used as the sensing element. Because voltage controlled inductors are rather large and have limited range, the natural choice is to use voltage controlled capacitors. As it turns out, all semiconductor junction devices have an effective capacitance across the junction. If a PN junction is doped specifically [TODO: update with more details + link to nanofab chapter/theory] with the intention of increasing the capacitance at the junction, then the device is called a varactor diode (or sometimes, a varicap).

However, using a varactor diode introduces some new complications, as diodes are inherently non-linear circuit components. As a result, there was some trial and error involved in identifying the best topology of varactor diode for differential operation that would minimize nonlinearities while attenuating common-mode noise. For the interested reader, several of these differential varactor topologies are presented for completeness in Appendix A for the interested reader. The next section will focus on impedance analysis in the laplace domain of

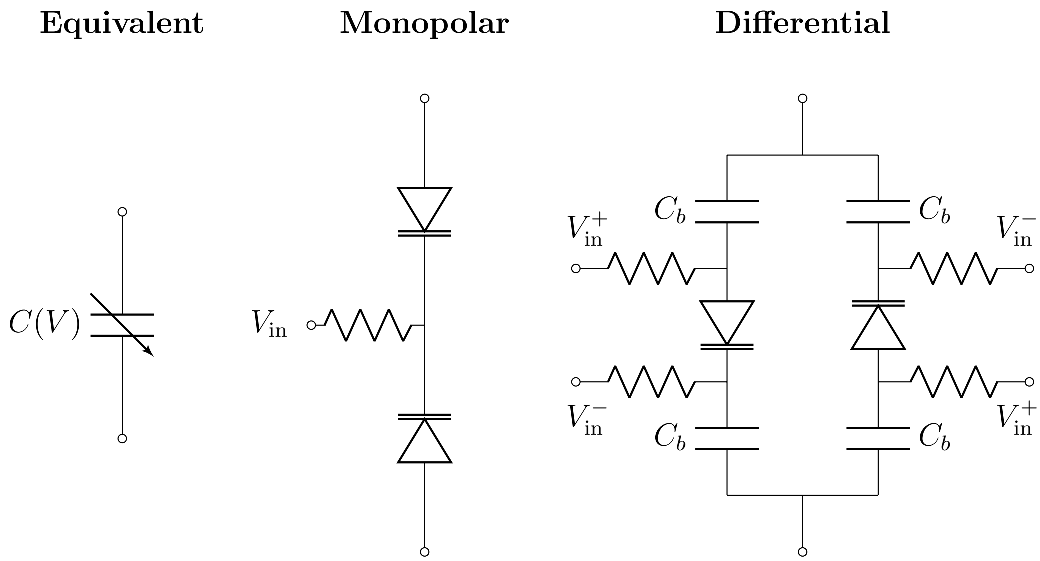


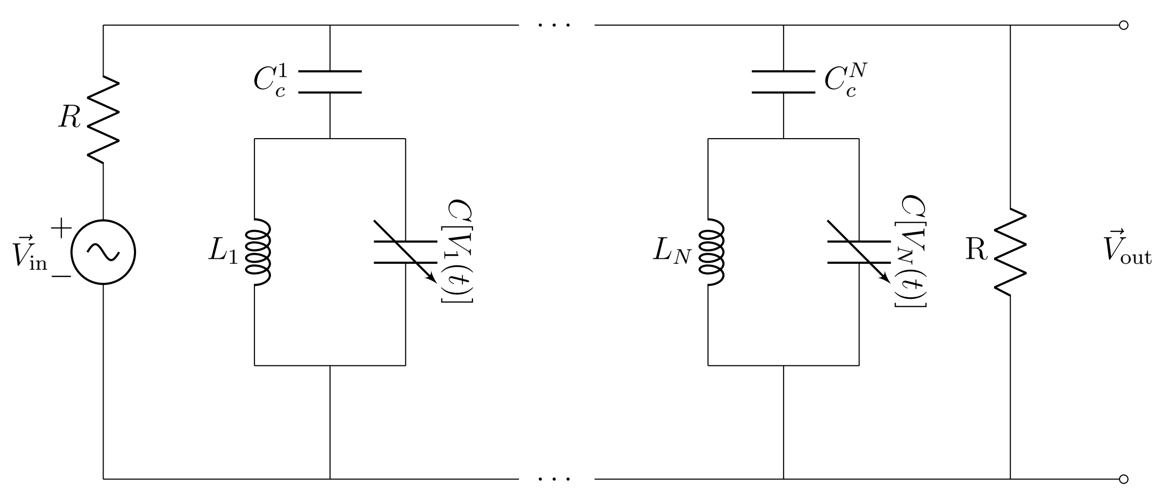
Figure 2.2: The monopolar and differential configurations can both be represented by the equivalent voltage controlled capacitor shown on the far left. Both configurations are intended as a stand-in for the capacitive element in an LC circuit. In order to be able to claim equivalence, all of the resistors must be large enough (on the order of MΩ) to decouple the voltage input and draw no current. The monopolar configuration is the standard use case for varactors in tuning circuits, and enables a range up to the breakdown voltage of the diode. The differential configuration has a voltage range limited by the requirement that the diode not enter the forward conducting mode. To keep the input voltage from mixing with the resonator sinusoid, blocking capacitors C_b are needed for this configuration. To ensure that the majority of the capacitance (and thus change in capacitance) comes from the varactor, the blocking capacitors should be at least an order of magnitude larger than the varactor zero-bias junction capacitance C_0.

an arbitrary set of parallel LC resonators on the same transmission line using the differential varactor configuration shown in Figure 2.2.

2.2 Impedance Analysis

The general idea is that each resonator will be responsible for encoding one signal. In the case of a capacitive sensing element (varactor diode), the voltage across the varactor diode is the signal being encoded. When there are multiple resonators in parallel, each resonator needs a capacitor to couple it to the transmission line. Though the capacitor serves to couple the LC resonators to the transmission line, it's main purpose is to isolate resonators from one another. Without it, all the resonator inductors and variable capacitors would be in parallel with each other, resulting in an equivalent circuit with only a single inductor and capacitor. Adding the coupling capacitor isolates all of the resonators, allowing N unique resonators to sit on the same transmission line.

The transmission line will be excited with a superposition of N signals, each intended to drive one of the N resonators at it's resonant frequency ω_i. As we'll see later, the depth of the resonance features encoding our signal is a function of the values of the inductor, variable capacitor, and the coupling capacitor. To understand how the circuit will respond, we can analyze the circuit's response to a single signal of frequency ω. To get the whole response, we can construct the response at each frequency ω_i, then sum them.



Because the signals we expect to record vary on timescales much slower than the time constant of the circuit, we analyze it as a steady state impedance network—our input signal will be a phasor with amplitude A_{in} and phase ϕ_{in} and our output signal will be a phasor with amplitude A_{out} and phase ϕ_{out}.

$$\vec{V}_{\text{in}} = A_{\text{in}} \exp\left[j(\omega t + \phi_{\text{in}})\right]$$
$$\vec{V}_{\text{out}} = A_{\text{out}} \exp\left[j(\omega t + \phi_{\text{out}})\right]$$

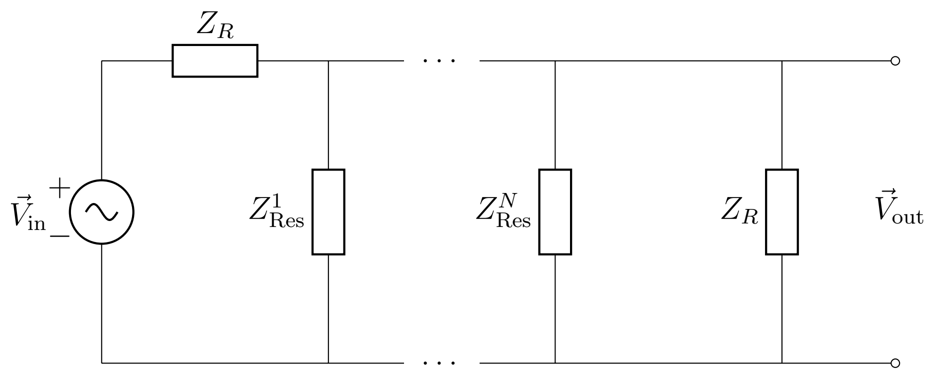
From $\vec{V}_{\text{in}}$ and $\vec{V}_{\text{out}}$, we can obtain a complex polar expression for the transfer function:

$$\vec{H} = \frac{\vec{V}_{\text{out}}}{\vec{V}_{\text{in}}}$$
$$\vec{H} = \frac{A_{\text{out}} \exp\left[j(\omega t + \phi_{\text{out}})\right]}{A_{\text{in}} \exp\left[j(\omega t + \phi_{\text{in}})\right]}$$
$$\vec{H} = \frac{A_{\text{out}}}{A_{\text{in}}} \exp\left[j(\phi_{\text{out}} - \phi_{\text{in}})\right]$$

The most important consequence of this notation is that the complex transfer function provides a relationship between the relative amplitude and phase of the output voltage as a function of the input phasor.

$$\vec{V}_{\text{out}} = \vec{H} \cdot \vec{V}_{\text{in}}$$
$$A_{\text{out}} = |\vec{H}| \cdot A_{\text{in}}$$
$$\phi_{\text{out}} = \angle \vec{H} + \phi_{\text{in}}$$

Because we can use impedances to write our circuit as a voltage divider, we can write our transfer function $\vec{H}$ as a function of complex impedances. Algebraically manipulating this to obtain an expression for the output amplitude will allow us to develop an analytical expression for how the circuit will respond. Analysis of this expression will give us insight into how to design the circuit for maximum effectiveness, as well as enable simulating the data we expect to produce as a function of the signals we expect to measure. With representative simulated data, we can validate a simulated readout system prior to implementing it in hardware.



By lumping each of the resonators into their own impedance Z^i_{Res}, it becomes much clearer that each of the resonators is in parallel with each other and Z_R. Using the rules for combining parallel and series networks of impedances, we can obtain an expression for the transfer function in terms of the impedances of our circuit network.

$$
\begin{aligned}
Z_{\text{tot}} &= Z_R + \left(\frac{1}{Z_R} + \frac{1}{Z^1_{\text{Res}}} + \ldots + \frac{1}{Z^N_{\text{Res}}}\right)^{-1} \\
Z_{\text{tot}} &= Z_R + \frac{1}{\frac{1}{Z_R} + \sum_{i=1}^{N} \frac{1}{Z^i_{\text{Res}}}} \\
Z_{\text{tot}} &= \frac{2 + Z_R \sum_{i=1}^{N} \frac{1}{Z^i_{\text{Res}}}}{\frac{1}{Z_R} + \sum_{i=1}^{N} \frac{1}{Z^i_{\text{Res}}}} \\
\frac{\vec{V}_{\text{out}}}{\vec{V}_{\text{in}}} &= \frac{\frac{1}{\frac{1}{Z_R} + \sum_{i=1}^{N} \frac{1}{Z^i_{\text{Res}}}}}{\frac{2 + Z_R \sum_{i=1}^{N} \frac{1}{Z^i_{\text{Res}}}}{\frac{1}{Z_R} + \sum_{i=1}^{N} \frac{1}{Z^i_{\text{Res}}}}} \\
\vec{H} &= \frac{1}{2 + Z_R \sum_{i=1}^{N} \frac{1}{Z^i_{\text{Res}}}}
\end{aligned}
$$

Now that we have an expression for the transfer function, we can work on simplifying it algebraically so that we can gain intuition about the circuit and obtain expressions for the relative phase and amplitude of our transmission wave. First, let's work out the impedance of a single resonator Z^i_{Res}.

$$Z^i_{\text{Res}} = Z^i_{C_c} + \left(\frac{1}{Z^i_L} + \frac{1}{Z_{C[V_i(t)]}}\right)^{-1}$$

$$Z^i_{\text{Res}} = \frac{1}{j\omega C^i_c} + \left(\frac{1}{j\omega L_i} + j\omega C[V_i(t)]\right)^{-1}$$

$$Z^i_{\text{Res}} = \frac{1}{j\omega C^i_c} + \left(\frac{1-\omega^2 L_i C[V_i(t)]}{j\omega L_i}\right)^{-1}$$

$$Z^i_{\text{Res}} = \frac{1}{j\omega C^i_c} + \frac{j\omega L_i}{1-\omega^2 L_i C[V_i(t)]}$$

$$Z^i_{\text{Res}} = \frac{1-\omega^2 L_i C[V_i(t)] - \omega^2 L_i C^i_c}{j\omega C^i_c(1-\omega^2 L_i C[V_i(t)])}$$

Plugging the impedances into our expression for the transfer function, we find

$$\vec{H} = \frac{1}{2 + j\displaystyle\sum_{i=1}^{N} \frac{R\omega C^i_c(1-\omega^2 L_i C[V_i(t)])}{1-\omega^2 L_i C[V_i(t)] - \omega^2 L_i C^i_c}}$$

Multiplying both the numerator and the denominator by the complex conjugate of the denominator, $\vec{H}$ becomes

$$\vec{H} = \frac{2 - j\displaystyle\sum_{i=1}^{N} \frac{R\omega C^i_c(1-\omega^2 L_i C[V_i(t)])}{1-\omega^2 L_i C[V_i(t)] - \omega^2 L_i C^i_c}}{(2)^2 + \left(\displaystyle\sum_{i=1}^{N} \frac{R\omega C^i_c(1-\omega^2 L_i C[V_i(t)])}{1-\omega^2 L_i C[V_i(t)] - \omega^2 L_i C^i_c}\right)^2}$$

$$\vec{H} = \frac{2}{4 + \left(\displaystyle\sum_{i=1}^{N} \frac{R\omega C^i_c(1-\omega^2 L_i C[V_i(t)])}{1-\omega^2 L_i C[V_i(t)] - \omega^2 L_i C^i_c}\right)^2} - j\frac{\displaystyle\sum_{i=1}^{N} \frac{R\omega C^i_c(1-\omega^2 L_i C[V_i(t)])}{1-\omega^2 L_i C[V_i(t)] - \omega^2 L_i C^i_c}}{4 + \left(\displaystyle\sum_{i=1}^{N} \frac{R\omega C^i_c(1-\omega^2 L_i C[V_i(t)])}{1-\omega^2 L_i C[V_i(t)] - \omega^2 L_i C^i_c}\right)^2}$$

Recalling that in complex space

$$|\vec{H}| = \sqrt{\text{Re}(\vec{H})^2 + \text{Im}(\vec{H})^2}$$

$$\angle\vec{H} = \arctan\left(\frac{\text{Im}(\vec{H})}{\text{Re}(\vec{H})}\right)$$

We can use our earlier expressions for A_{out} and ϕ_{out} to obtain an expression for the relative amplitude and phase of the output phasor.

$$|\vec{H}| = \frac{1}{\sqrt{4 + \left(\sum_{i=1}^{N} \frac{R\omega(C_t^i + C[V_i(t)])}{1 - \omega^2 L_i C[V_i(t)] - \omega^2 L_i C_c^i} \right)^2}} \tag{2.1}$$

$$\angle\vec{H} = \arctan \left(\frac{-\sum_{i=1}^{N} \frac{R\omega(C_t^i + C[V_i(t)])}{1 - \omega^2 L_i C[V_i(t)] - \omega^2 L_i C_c^i}}{2} \right) \tag{2.2}$$

2.3 Frequency Response

Armed with our expression for the transfer function of the input phasor to the output phasor, we can characterize the amplitude and phase behavior of a resonator around the frequency it resonates at. Note that these functions are functions of two variables—the frequency ω of the sinusoid on the transmission line, and the voltage $V_i(t)$ applied across the varactor diode. During operation of the sensors, the frequency of the sinusoid on the transmission line is constant, and each resonator is being driven at it's own unique resonant frequency.

However, to get intuition for resonator frequency domain bandwidth and response, it is useful to instead analyze the response of the circuit over a frequency sweep while the voltage across the varactor is constant. For simplicity, we focus on analyzing the frequency response of the i^{th} resonator, and so long as we confine our analysis to the frequency band containing this particular resonator we can safely ignore the effects of other resonators and drop the summations from our equation. In that case we excite the transmission line with a single frequency ω which sweeps the frequency band at an instant in time t_0 such that $V_i(t) = 0$, and thus $C[V_i(t) = 0] = C_0$. Therefore, we can rewrite our expressions for amplitude and phase from the previous section as follows:

$$|\vec{H}(\omega, V_i(t_0) = 0)| = \frac{1}{\sqrt{4 + \left(\frac{R\omega C_c^i(1-\omega^2 L_i C_0)}{1-\omega^2 L_i C_0 - \omega^2 L_i C_c^i}\right)^2}}$$

$$\angle\vec{H}(\omega, V_i(t_0) = 0) = \arctan\left(-\frac{R\omega C_c^i(1-\omega^2 L_i C_0)}{2(1-\omega^2 L_i C_0 - \omega^2 L_i C_c^i)}\right)$$

Though the next section will focus on how to select the coupling capacitance and resonator inductance given the transmission line resistance, varactor diode voltage-capacitance response, and resonance frequency, for now we will just assume that these values are given and show the results of our analytical expressions. We'll also assume that the input phasor has unit amplitude and zero phase to facilitate visualization of the order of the effect of the resonant circuit on these parameters.

2.3.1 Amplitude Response

The amplitude response of the circuit is straightforward. With a varactor diode whose zero bias-voltage capacitance C_0 is 25 pF, a transmission line of resistance R of 1 kΩ, a coupling capacitance C_c^i of ∼33 fF, and a ∼4 μH inductance L_i, we can form a resonator with the following frequency response.

These particular resonator component values have been selected to ensure that the resonant frequency of the resonator is at 50 Mhz. The relative amplitude of the output phasor as compared to the input phasor drops sharply around the resonant frequency of the resonator, but is mostly unaffected at frequencies away from that resonant frequency. This is a desired property which enables frequency domain multiplexing—so long as each resonator has it's own band of frequencies which are not encroached upon by other resonators, it won't interfere with other resonators on the same transmission line.

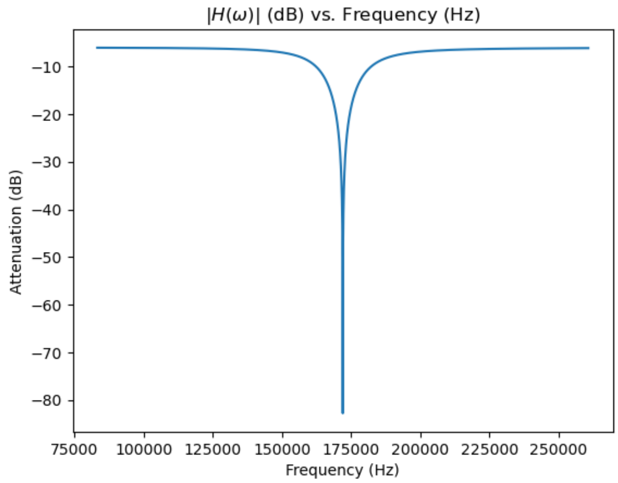


Figure 2.3: The relative amplitude of the output phasor as compared to an input phasor with zero phase and unit amplitude applied to the transmission line of a resonator with the following parameters: zero bias-voltage capacitance C_0 of 25 pF, a transmission line of resistance R of 1 kΩ, a coupling capacitance C_c^i of $\sim$33 fF, and a $\sim$4 μH inductance L_i.

2.3.2 Phase Response

For the sake of simplicity, we'll use the same resonator parameters (zero bias-voltage capacitance C_0 of 25 pF, a transmission line of resistance R of 1 kΩ, a coupling capacitance C_c^i of $\sim$33 fF, and a $\sim$4 μH inductance L_i) to plot the phase frequency response. However, it's a little bit more complicated to plot the relative phase of the output phasor as compared to the input phasor because the expression involves an arctangent of the imaginary and real parts of the transfer function. However, the period of the tangent function is only from $-\pi/2$ to $\pi/2$, which causes a couple problems when we take the arctangent of phase. The first and most significant problem is that phase is periodic—when we add any multiple of 2π to it, we should get the same phase. The second problem is that with our current analytical expression for the phase, it happens that there is a discontinuity in phase as the frequency sweep passes over the resonant frequency.

The first problem can be resolved by redefining our expression for phase to ensure that

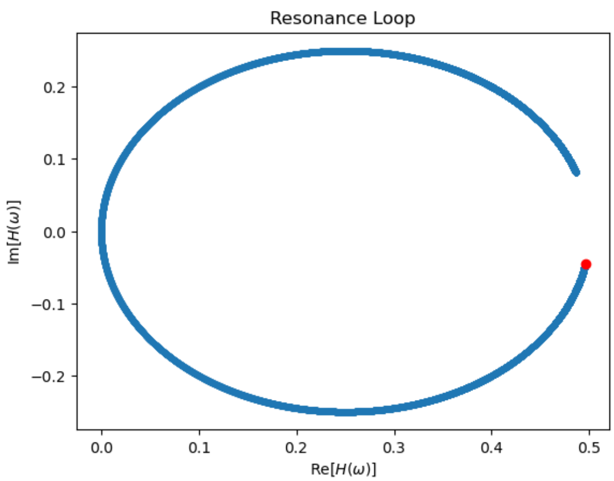


Figure 2.4: The transfer function $\vec{H}$ of a resonator with the following parameters: zero bias-voltage capacitance C_0 is 25 pF, a transmission line of resistance R of 1 kΩ, a coupling capacitance C_c^i of $\sim$33 fF, and a $\sim$4 μH inductance L_i. The red dot indicates the location in the complex plane at the lowest frequency in the frequency sweep.

it has a range of 2π, and the second problem can be resolved by phase wrapping—modolo division of each point by 2π and keeping the remainder will eliminate any discontinuities that arise from crossing over the resonant frequency. To motivate a solution for how to redefine our expression of phase, we plot the transfer function $\vec{H}$ in the complex plane.

From this figure, we can see that a much more natural definition of phase would be to identify the center of the loop that the transfer function sweeps out during a frequency sweep. Then, we can re-define phase $\angle\vec{H} \to \angle\vec{H}'$ as the angle swept out by a phasor centered at this loop (which notably, is not centered at the origin because the real part of the transfer function is positive definite). Finally, we define zero phase as the angle made by this new phasor when it points to the location of the transfer function in the complex plane corresponding to the resonant frequency of the resonator. This gives our relative phase the expected range of 2π while enabling both negative and positive phase shifts.

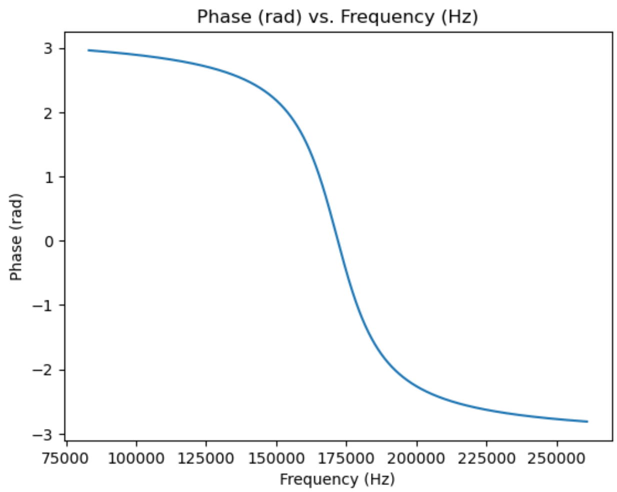


Figure 2.5: Relabeled phase $\angle \vec{H}'$ of the output phasor with respect to the input phasor, from a resonator with the following parameters: zero bias-voltage capacitance C_0 of 25 pF, a transmission line of resistance R of 1 kΩ, a coupling capacitance C_c^i of ∼33 fF, and a ∼4 μH inductance L_i.

2.4 Resonator Sensitivity & Bandwidth

Now, with the intuition we've gained for how a resonator responds to a frequency sweep, we can focus on analytically determining how to select the coupling capacitance C_c^i and resonator inductance L_i given the transmission line resistance, varactor diode voltage-capacitance response, and desired resonant frequency. Since different varactor diodes have different voltage-capacitance responses, we'll need to work backwards to determine the selectivity of the resonator.

Suppose we have a varactor diode whose capacitance with zero bias voltage is C_0. Then, as the diode is biased, this capacitance will change to some new value, $C_0 + \Delta C(V)$. In other words, suppose the signal we'd like to measure has a maximum amplitude of $V_{\max}$. When this voltage is applied across the varactor, it will result in a corresponding maximum change in capacitance $\Delta C_{\max}$. Note that most varactor diode datasheets' voltage-capacitance curves do not have fine enough resolution to show capacitances resulting from changes on the order of millivolts, which means it will be necessary to characterize the varactor diode's voltage-capacitance response.

To ensure there is sufficient sensitivity in phase to capture these changes in capacitance (and thus, capture these changes in voltage), we need to ensure that our shifts in phase fall within the linear region of the our phase response from Figure A.6. We could accomplish this by working with an expression of the phase, but since that involves an arctangent it gets algebraically messy pretty quickly. Also, there's not a clearly defined cutoff between the linear and asymptotic regions of the phase response, so preferably we'd like a meaningful way to select that arbitrary cutoff point. To do this, we return to equation 2.1 and our expression for the relative amplitude of the output phasor. By picking resonator components so that the full-width half-maximum (FWHM) of the resonance feature corresponds with the maximum change in capacitance, we can be confident that the phase response will fall within the linear region.

To be able to solve for the coupling capacitance and inductance, we'll need a system of two equations. Our first equation will be an expression for the resonant frequency of a resonator, and the second equation will come from our desired resonator sensitivity. To get the first equation, we'll need to write down equation 2.1 for a single resonator, recalling that we excite the transmission line with a sinusoid of constant frequency ω_0^i corresponding to resonance at zero varactor bias voltage. For notational ease, we will drop all the i subscripts and endeavor to remember that every value except for C_0 and R are unique to the resonator in the analysis.

$$|\vec{H}(\omega = \omega_0, V(t_0) = 0)| = \frac{1}{\sqrt{4 + \left(\frac{R\omega_0 C_c(1-\omega_0^2 LC_0)}{1-\omega_0^2 LC_0 - \omega_0^2 LC_c}\right)^2}}$$

Recalling our earlier impedance analysis, note that the output phasor amplitude is $\propto \frac{1}{Z^i_{\text{res}}}$. The physical interpretation of this type of resonator at resonant frequency is that the impedance of the resonator drops to zero, causing the entire resonator to act like a short between the transmission line and the ground for signals at that frequency. Shorting the transmission to ground before the point at which the output phasor is measured effectively makes the amplitude of the output phasor equal to zero. Thus, to get our first equation relating the resonant frequency to the amplitude of the output phasor, we set the above expression equal to zero. Since the numerator is positive definite, the only possible way for $\vec{H}$ to become zero is in the limit in which the denominator approaches infinity. In order for the denominator of $\vec{H}$ to be equal to infinity, then

$$\begin{aligned} 0 &= 1 - \omega_0^2 L C_0 - \omega_0^2 L C_c \\ \omega_0^2 &= \frac{1}{L(C_0 + C_c)} \end{aligned}$$

Notably, we recover the expression for the resonant frequency of a standard tank resonator, $\omega_0^2 = \frac{1}{LC}$, with the capacitance of the varactor diode and the coupling capacitor added together. Now that we have an expression for the resonant frequency, we can focus on obtaining an expression for resonator sensitivity to a change in capacitance ΔC_{max} using equation 2.1 when $V(t) = V_{\mathrm{max}}$. The maximum possible value of $\vec{H}$ is 1/2, so then the FWHM of the resonance feature corresponds with when $\vec{H}$ is 1/4.

$$\begin{aligned} |\vec{H}(\omega = \omega_0, V(t_0) = V_{\mathrm{max}})| &= \frac{1}{\sqrt{4 + \left(\frac{R\omega_0 C_c(1 - \omega_0^2 L[C_0 + \Delta C_{\mathrm{max}}])}{1 - \omega_0^2 L[C_0 + \Delta C_{\mathrm{max}}] - \omega_0^2 L C_c}\right)^2}} \\ \frac{1}{4} &= \frac{1}{\sqrt{4 + \left(\frac{R\omega_0 C_c(1 - \omega_0^2 L[C_0 + \Delta C_{\mathrm{max}}])}{1 - \omega_0^2 L[C_0 + \Delta C_{\mathrm{max}}] - \omega_0^2 L C_c}\right)^2}} \\ 16 &= 4 + \left(\frac{R\omega_0 C_c(1 - \omega_0^2 L[C_0 + \Delta C_{\mathrm{max}}])}{1 - \omega_0^2 L[C_0 + \Delta C_{\mathrm{max}}] - \omega_0^2 L C_c}\right)^2 \\ \sqrt{12} &= \frac{R\omega_0 C_c(1 - \omega_0^2 L[C_0 + \Delta C_{\mathrm{max}}])}{1 - \omega_0^2 L[C_0 + \Delta C_{\mathrm{max}}] - \omega_0^2 L C_c} \end{aligned}$$

We've now written down two equations for our resonators in terms of constants (R, C_0), parameters we determine during the design process of a resonator (desired resonant frequency ω_0 and maximum change in capacitance ΔC_{max}), and the resonator specific values we'd like to solve for (L, C_C). By re-writing our first equation for the resonant frequency to get L in terms of C_c, we can substitute this into our second equation and solve for C_c.

$$\begin{aligned}
L &= \frac{1}{\omega_0^2(C_0 + C_c)} \\
\sqrt{12} &= \frac{R\omega_0 C_c(1 - \omega_0^2 L[C_0 + \Delta C_{\max}])}{1 - \omega_0^2 L[C_0 + \Delta C_{\max}] - \omega_0^2 L C_c} \\
\sqrt{12} &= \frac{R\omega_0 C_c(1 - \frac{C_0 + \Delta C_{\max}}{C_0 + C_c})}{1 - \frac{C_0 + \Delta C_{\max} + C_c}{C_0 + C_c}} \\
\sqrt{12} &= \frac{R\omega_0 C_c(C_0 + C_c - [C_0 + \Delta C_{\max}])}{C_0 + C_C - (C_0 + \Delta C_{\max} + C_c)} \\
\sqrt{12} &= -\frac{R\omega_0 C_c}{\Delta C_{\max}}(C_c - \Delta C_{\max}) \\
0 &= \frac{R\omega_0}{\Delta C_{\max}} C_c^2 - R\omega_0 C_c + \sqrt{12}
\end{aligned}$$

Solving this quadratic for C_c and plugging it into our expression for the inductance gives us the following:

$$C_c = \frac{\Delta C_{\max}}{2} + \sqrt{\frac{\Delta C_{\max}^2}{4} + \frac{\sqrt{12}\Delta C_{\max}}{R\omega_0}} \tag{2.3}$$

$$L = \frac{1}{\omega_0^2(C_c + C_0)} \tag{2.4}$$

To calculate how much bandwidth the resonator is going to consume under these design specifications, we can plug in our calculated values for C_c and L into our expression for $\vec{H}$, then solve for the frequency at which the FWHM occurs while no voltage is applied across the varactor diode. Since we already know the resonant frequency, the solution will give us a conservative estimate to how much frequency bandwidth the resonator will consume.

$$\frac{1}{4} = \frac{1}{\sqrt{4 + \left(\frac{R\omega C_c(1 - \omega^2 LC_0}{1 - \omega^2 LC_0 - \omega^2 LC_c}\right)^2}}$$

$$16 = 4 + \left(\frac{R\omega C_c(1 - \omega^2 LC_0}{1 - \omega^2 LC_0 - \omega^2 LC_c}\right)^2$$

$$\sqrt{12} = \frac{R\omega C_c(1 - \omega^2 LC_0)}{1 - \omega^2 LC_0 - \omega^2 LC_c}$$

$$0 = (-RLC_0C_c)\omega^3 + (LC_0\sqrt{12} + LC_c\sqrt{12})\omega^2 + (RC_c)\omega - \sqrt{12}$$

Unfortunately, the roots to this cubic do not have a nice closed-form solution. To find them, we need to perform the change of variable $\omega = t - b/3a$, which will reduce the cubic equation $a\omega^3 + b\omega^2 + c\omega + d$ to it's depressed form $t^3 + pt + q$, where

$$p = \frac{3ac - b^2}{3a^2}$$

$$q = \frac{2b^3 - 9abc + 27a^2d}{27a^3}$$

Using Vietè's formula for the roots of a depressed cubic, the solutions to the equation $t^3 + pt + q = 0$ are given by

$$t_k = 2\sqrt{-\frac{p}{3}}\cos\left[\frac{1}{3}\arccos\left(\frac{3q}{2p}\sqrt{-\frac{3}{p}}\right) - \frac{2\pi k}{3}\right] \quad \text{for } k = 0, 1, 2$$

To recover the roots of the cubic we care about, we return to our expression for the change of variables.

$$\omega_k = t_k - \frac{b}{3a}$$

Because this is a cubic equation, there will be three roots. The root with the largest absolute value can be discarded, as it is not related to the resonance dip and is instead a function of the overall shape of the curve as seen in figure: [TO DO: MAKE PLOT COMPARING $|H|$ TO $1/\sqrt{1 + x^2}$]. In practice, this means we discard the root corresponding to $k = 0$. To get the $\Delta\omega_{\text{FWHM}}$, we take the absolute value of the remaining two roots and subtract them from each other:

$$|\Delta\omega_{\text{FWHM}}| = \big|\,|\omega_{\pm}| - |\omega_{\mp}|\,\big|$$

2.5 Responsivity

[analyze how much the amplitude and phase are affected by a x% change in capacitance, see bmazin thesis section 2.4]

2.6 Coupling Capacitance

[Spice simulations of the circuit? At the very least outline where coupling capacitors would go and note value discrepancy to avoid lopsidedness/attenuation (still open question). High frequency rolloff implies I may want to consider looking into a circulator]

CHAPTER 3

Digital Signal Processing

Write a paragraph to transition from previous chapter about circuit theory to this chapter about the digital signal processing theory that goes into exciting the resonators and reading out the channels

3.1 Resonator Excitation

Though there are several different ways to generate the signal to drive all of the resonators on the transmission line of the detector circuit, the premise is the same: For every resonator, generate a sinusoid oscillating at the resonant frequency of the aforementioned resonator. Summing all of these sinusoids together gives a single time-domain voltage signal which can be applied to the transmission line. The output voltage is this same time-domain signal, but each individual sinusoid in the summation will be modified by the amplitude and phase that are calculated in Section 2.1.1. This amplitude and phase contain information about the voltage applied to the variable capacitive element and are the primary means by which this signal can be extracted.

One method (TODO: make bib + cite mcHugh 2012, van rantjwick 2016, strader 2016) of generating this excitation signal is to start with superposition of complex I/Q signals, with one I (In-phase) and Q (Quadrature) component per channel i.

$$\begin{aligned} V_{\text{in}} &= \sum_i^N I(t)_i + jQ(t)_i \\ &= \sum_i^N \cos(\omega_i t) + j\sin(\omega_i t) \end{aligned}$$

The downside of this approach is that in order to represent this complex waveform as a real signal, the cosine and sine terms need to be produced independently and combined in hardware with an IQ mixer to produce the complete time domain input signal to be passed through the resonators. Not only that, but to then extract the phase of each channel after passing through the transmission lines, the output signal has to be separated back into I and Q and then sampled and digitized. This complexity is somewhat justified by the fact that the IQ signal is what makes it tractable to extract the phase in the first place, but it leaves much to be desired.

As an alternative, I propose to start with a real signal and produce quadrature components during digital signal processing, which has the benefit of simplifying the hardware and—as I will show in the next section—improving the time resolution of our extracted phase when performing digital signal processing on the output signal. This is of particular note because the algorithm presented in (TODO: cite van rantjwick 2016) consists of a large ($2^{14} \sim 2^{16}$ points) fourier transform (FFT) to ensure that no resonant frequency falls into the side-lobes in-between bins in frequency space. A notable downside of this simple approach is that it significantly limits time resolution—each FFT window corresponds to a single measurement of phase per channel, so the larger the FFT the less often the phase is sampled.

To overcome this limitation, Strader 2016 (TODO: citation needed) uses much smaller (2^{11} point) FFTs, but this necessitates additional complexity in the digital signal processing to widen the spectral bin passband, divide bins into chunks (each of which may contain zero, one, or two channels) to isolate individual channels, as well as requires a scheme in which time-domain samples are passed into two different FFTs offset from one another by half the window size and the outputs are interleaved into a coherent timestream with double the time resolution. Each FFT window still provides only a single measurement of a phase per channel—but time resolution is significantly improved!

However, by starting a with a real time-domain signal instead of a complex time-domain signal signal, it is possible to produce a representation of the phase in each channel with a one-to-one correlation of each phase measurement to a sample of the output signal in the time domain. To accomplish this, our input signal is the real sinusoid

$$V_{\text{in}} = \sum_{i}^{N} \sin(\omega_i t)$$

and our output signal will be the real sinusoid

$$V_{\text{out}} = \sum_{i}^{N} A_i[t] \sin(\omega_i t + \phi_i[t])$$

3.2 DSP Algorithm

Digital Signal Processing Algorithm

$V(k) \leftarrow \mathcal{F}\{V(t)\}$ ▷ Apply to both $V_{\text{in}}(t)$ and $V_{\text{out}}(t)$ in parallel

$V_A(k) \leftarrow 2U(k)V(k)$ ▷ $U(k)$ is the Heaviside step function

Duplicate $V_A(k)$ N times ▷ N is the number of channels

for $i \in 0, 1, \cdots, N$ **do**

  $V_{A_i}(k') \leftarrow V_{A_i}(k) * \mathcal{F}\{\exp(-j\omega_i t)\}$

  $V_{A_i}(t) \leftarrow \mathcal{F}^{-1}\{V_{A_i}(k')\}$

  $V_{A_i}(t') \leftarrow G(\omega) * V_{A_i}(t)$ ▷ $G(\omega)$ is a lowpass filter

  $H_i(t) \leftarrow \frac{V_{\text{out}}(t')_i}{V_{\text{in}}(t')_i}$

  $\phi_i(t) \leftarrow \arctan\left(\frac{\text{Im}[H_i(t)]}{\text{Re}[H_i(t)]}\right)$

end for

(TODO: GENERATE FIGURES/visualizations for algorithm steps)

At a high level, the goal of this algorithm is to take a real, 'In-phase' (In-phase being a little misleading in this case on account of there being no Quadrature component) time-domain signal and construct from it a Quadrature component. Then, this signal is decomposed so as to isolate individual channels, which are finally converted into a real, time-domain measurement of the phase of each resonator. Broadly, the algorithm can be broken down into three parts: Steps numbered 1-2 construct the quadrature component of the signal, steps 3-7 isolate individual channels, and steps 8-10 extract the phase.

Finally, because our signal is in the time domain, we pass overlapping chunks of time-domain signal to the algorithm and stitch together the output so as to avoid windowing effects (similar to the overlap-save method for fast convolution). In simulation, using an overlap ratio of 1:8 on each side of the window worked extremely well. That is to say, for real-time processing, one would pick an FFT size of M points based on system constraints/considerations, sequentially sample $1/8M + M + 1/8M$ points, apply the algorithm to this window, and then collect the next sample of points. To generate a coherent time stream, each window interface will ultimately overlap by $1/4M$. (TODO: generate really nice figure using software from Dean's teaching session detailing the windowing procedure for real-time processing)

3.2.1 Quadrature Construction

Since we have access to the carrier sinusoid (we are producing it, after all!), we can extract the phase from our I/Q data by determining what proportion of the output signal is in-phase and in-quadrature with the carrier. This can be achieved by producing an in-quadrature signal for the carrier(the input sinusoid) and for the output signal—in other words a 90° phase shift. Because the entire algorithm is operating on digital data, and because this data is real valued, this can be achieved trivially by producing the Analytic signal of the carrier and output signals. The fourier transform of a real signal has a peak in frequency on both the negative and positive side of the spectrum—by multiplying the frequency domain representation of our sinusoids with the heaviside step function, we eliminate the negative frequencies and produce the spectrum corresponding to a frequency domain representation of our desired I/Q signal. In other words, if we were to perform steps 1 & 2 of the outlined algorithm and then take the inverse fourier transform of our signal, our result would be the following:

$$V_{\text{in}}(t) \rightarrow \sum_{i}^{N}[\sin(\omega_i t) + j\cos(\omega_i t)]$$

$$V_{\text{out}}(t) \rightarrow \sum_{i}^{N} A_i[t][\sin(\omega_i t + \phi_i[t]) + j\cos(\omega_i t + \phi_i[t])]$$

Note that with Euler's identity, this becomes

$$V_{\text{in}}(t) \rightarrow \sum_{i}^{N}\exp(j\omega_i t)$$

$$V_{\text{out}}(t) \rightarrow \sum_{i}^{N} A_i[t]\exp(j\{\omega_i t + \phi_i[t]\})$$

3.2.2 Downconversion and Channel Isolation

On it's own, this result is useless without a way of channelizing. Operating under the assumption that one has an IQ waveform in the time-domain, one possible approach to channelization suggested in Strader 2016 (TODO: citation) is to duplicate said time-domain signal by the number of channels, downshift the frequency spectrum so that the frequency for the ith channel is at zero, and then lowpass filter to isolate that channel. In order to downshift the frequency spectrum, the ith duplicate is multiplied in the time-domain by the complex sinusoid $\exp(-j\omega_i t)$.

$$\begin{aligned}
V(t) \cdot \exp(-j\omega_k t) &= \sum_i^N \exp(j\omega_i t + \phi_i[t]) \cdot \exp(-j\omega_k t) \\
&= \sum_i^N \exp(j[\omega_i - \omega_k]t + \phi_i[t]) \\
&= \exp(\phi_k[t]) + \sum_{i \neq k}^N \exp(j[\omega_i - \omega_k]t + \phi_i[t])
\end{aligned}$$

Strader discards this suggestion as infeasible due to memory constraints because that downconversion requires N (channel number) pointwise multiplications of M (FFT length) points, requiring the storage of N lookup tables in memory (one for each complex waveform), and N multiplies of size M being carried out in parallel. To overcome these limitations, the downshift can happen in the frequency domain—which theoretically only requires the storage of M FFT coefficients per complex sinusoid rather than a full look-up table, and in practice requires no storage of any memory on lookup tables, whatsoever.

Theoretically speaking, we can avoid the troublesome pointwise multiplication and requisite lookup tables by performing an FFT of our complex waveform and convolving it with our frequency space representation of our desired channel's analytic signal. Because the convolution theorem states that a convolution in the frequency-domain is identical to a multiply in the time-domain, we achieve our desired effect. If we were to implement this step as proposed, we would effectively trade memory for computational complexity. However, we know apriori what the frequency domain representation of this convolution should look like—the entire frequency representation should shift so that the bin corresponding to the channel of interest should be centered at zero. So, in practice we can just do this manually rather than bothering to calculate a convolution.

After taking the IFFT of our downshifted signal, we can apply a low-pass filter with a cutoff frequency corresponding to our channel bandwidth, which (working under the assumption that the channels have been spaced appropriately) will isolate the channel of interest.

The end result of this is that we should now have N channels where the ith channel consists of the following complex waveform:

$$V_{\text{in}}(t')_i \rightarrow \exp(j\omega_i' t)$$
$$V_{\text{out}}(t')_i \rightarrow A_i[t]\exp(j\{\omega_i' t + \phi_i[t]\})$$

Technically, we would expect ω_i' to be roughly equal to zero, but because the complex sinusoid frequency may not line up with the center of a frequency bin (especially as the FFT window size decreases), in practice it is slightly offset and thus still plays a role in phase extraction.

3.2.3 Phase Extraction

Finally, to extract the phase we divide the transformed output signal $V_{\text{out}}(t')_i$ by the transformed carrier wave $V_{\text{in}}(t')_i$ to generate the transfer function for the resonator as discussed in section 2.1.1.

$$\begin{aligned} H_i(t) &= \frac{V_{\text{out}}(t')_i}{V_{\text{in}}(t')_i} \\ &= \frac{A_i[t]\exp(j\{\omega_i' t + \phi_i[t]\})}{\exp(j\omega_i' t)} \\ &= A_i[t]\exp(j\phi_i[t]) \\ &= A_i[t][\cos(\phi_i[t]) + j\sin(\phi_i[t])] \end{aligned}$$

Thus, it becomes clear that the phase can be extracted by taking the arctangent of the imaginary and real components of this transfer function.

$$\begin{aligned} \phi_i(t) &= \arctan\left(\frac{\text{Im}[H_i(t)]}{\text{Re}[H_i(t)]}\right) \\ &= \arctan\left(\frac{A_i(t)\sin(\phi_i[t])}{A_i(t)\cos(\phi_i[t])}\right) \end{aligned}$$

In practice, to account for phase unwrapping this output actually becomes $\phi_i(t)\%2\pi$, and to offset our phase calculation to be centered at the loop in the complex plane, the complex coordinates of that center point need to be subtracted from the imaginary and real arguments in the arctangent function, respectively.

3.2.4 Simulated Results

To demonstrate a proof of concept of this algorithm working in simulation, I developed a simulated circuit and randomly generate an underlying signal that the resonator is 'detecting'. After converting this expected measurement into the expected phase and amplitude as outlined in Chapter 2, I produce a simulated carrier wave and simulated output. Only the simulated carrier wave, simulated output, and frequency location/bandwidth of the channels are provided to the digital signal processing algorithm.

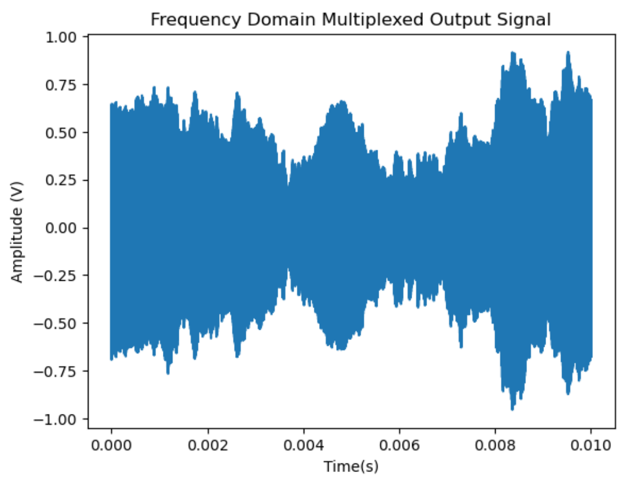


Figure 3.1:

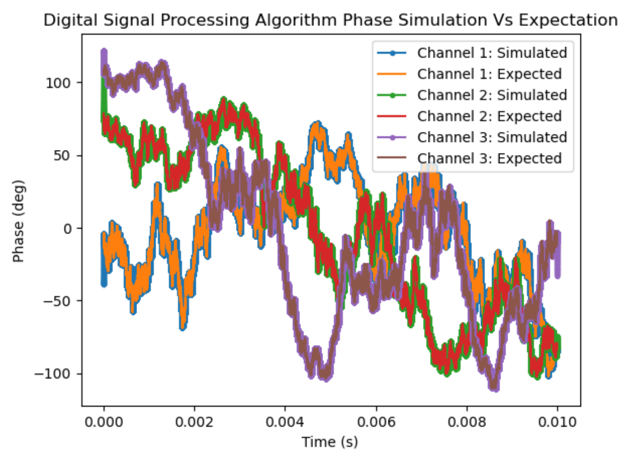


Figure 3.2:

The simulated output of a superposition of 51 MHz, 52 Mhz, and 53 MHz carrier waves after being modulated by the resonators is shown in Figure 3.1. Since the simulated measurement at each resonator is used to generate the simulated output wave and therefore is known, we have a benchmark with which to compare the results of the proposed digital signal processing algorithm. Because the results shown in Figure 3.2 would be indistinguishable otherwise, the simulated phase is plotted using markers, while the expected phase is plotted on top of the simulated phase without markers. All three channels are plotted simultaneously to demonstrate that the algorithm is successful in decoupling them.

NEXT STEPS: hardware development, add white noise and see how performance degrades

APPENDIX A

Alternate Topologies

A.1 Differential Varactor Topologies

[TODO: ADD SECTION TEXT]

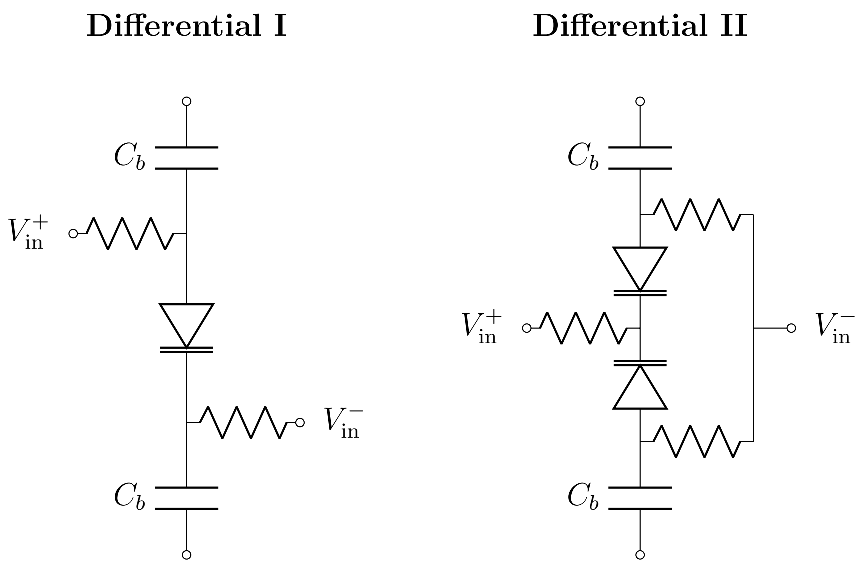


Figure A.1: Presented here are a few differential topologies of varactor diodes that introduced fairly large nonlinearities into the transmission line (Differential I) or did not adequately (for our application) attenuate common mode noise (Differential II).

A.2 Alternate Resonator Configurations

[TODO: - add all LC circuits in parallel with one another but each resonator is in series - all LC circuits in series with one another but each resonator is in parallel (all in transmission line)]

A.3 Impedance Analysis of Series LC Topology

[TODO: Update language to reflect that this is now in the appendix]

As alluded to in the previous sections, each resonator will be responsible for encoding one differential voltage signal through an interface with the body (usually electrodes). The transmission line will be excited with a superposition of N signals, each intended to drive one of the N resonators at it's resonant frequency ω_i. To understand how the circuit will respond, we can analyze the circuit's steady-state impedance response to a single signal of frequency ω. To construct the whole response, we can calculate the response at each resonant frequency ω_i, then sum all of the responses together.

As mentioned in the introduction to this chapter, the signals we expect to record vary on timescales much slower than the time constant of the circuit, so we analyze it as a steady state impedance network. Our input signal will be a phasor with amplitude A_{in} and phase ϕ_{in} and our output signal will be a phasor with amplitude A_{out} and phase ϕ_{out}.

$$\vec{V}_{\text{in}} = A_{\text{in}} \exp\left[j(\omega t + \phi_{\text{in}})\right]$$
$$\vec{V}_{\text{out}} = A_{\text{out}} \exp\left[j(\omega t + \phi_{\text{out}})\right]$$

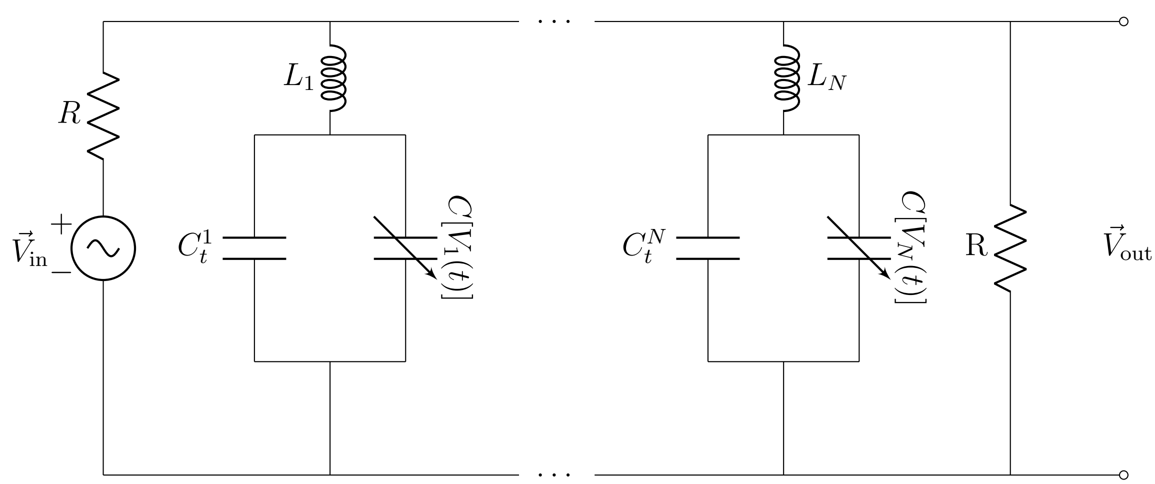


Figure A.2: This circuit demonstrates a configuration in which N resonators can be coupled onto the same transmission line, as well as indicates the location in which the input sinusoids should be applied. The variable capacitor is a stand-in for the differential configuration shown in Figure 2.2. The tuning capacitor C_t is not strictly necessary for the resonator to function, but as the ensuing analysis shows, can be used in tandem with the value of the inductor to fix the zero-bias resonant frequency and resonator bandwidth.

From $\vec{V}_{\text{in}}$ and $\vec{V}_{\text{out}}$, we can obtain a complex polar expression for the transfer function:

$$\vec{H} = \frac{\vec{V}_{\text{out}}}{\vec{V}_{\text{in}}}$$
$$\vec{H} = \frac{A_{\text{out}} \exp\left[j(\omega t + \phi_{\text{out}})\right]}{A_{\text{in}} \exp\left[j(\omega t + \phi_{\text{in}})\right]}$$
$$\vec{H} = \frac{A_{\text{out}}}{A_{\text{in}}} \exp\left[j(\phi_{\text{out}} - \phi_{\text{in}})\right]$$

The most important consequence of this notation is that the complex transfer function provides a relationship between the relative amplitude and phase of the output voltage as a function of the input phasor.

$$\vec{V}_{\text{out}} = \vec{H} \cdot \vec{V}_{\text{in}}$$
$$A_{\text{out}} = |\vec{H}| \cdot A_{\text{in}}$$
$$\phi_{\text{out}} = \angle\vec{H} + \phi_{\text{in}}$$

Because we can use impedances to write our circuit as a voltage divider, we can write our transfer function $\vec{H}$ as a function of complex impedances. Algebraically manipulating this to obtain an expression for the output amplitude will allow us to develop an analytical expression for how the circuit will respond. Analysis of this expression will give us insight into how to design the circuit for maximum effectiveness, as well as enable simulating the data we expect to produce as a function of the signals we expect to measure. With representative simulated data, we can validate a simulated readout system prior to implementing it in hardware. The first step is working out the impedance of a single resonator Z^i_{Res}.

$$\begin{aligned}
Z^i_{\text{Res}} &= Z^i_L + \left(\frac{1}{Z^i_{C_t}} + \frac{1}{Z_{C[V_i(t)]}}\right)^{-1} \\
Z^i_{\text{Res}} &= j\omega L_i + \left(j\omega C^i_t + j\omega C[V_i(t)]\right)^{-1} \\
Z^i_{\text{Res}} &= j\omega L_i + \frac{1}{j\omega C^i_t + j\omega C[V_i(t)]} \\
Z^i_{\text{Res}} &= \frac{1 - \omega^2 L_i C^i_t - \omega^2 L_i C[V_i(t)]}{j\omega C^i_t + j\omega C[V_i(t)]}
\end{aligned}$$

By lumping each of the resonators into their own impedance Z^i_{Res} as shown in Figure A.3, it becomes much clearer that the impedance of the final resistor Z_R and all of the impedances of the resonators share the same nodes and are thus in parallel.

Using the rules for combining parallel and series networks of impedances, we can obtain

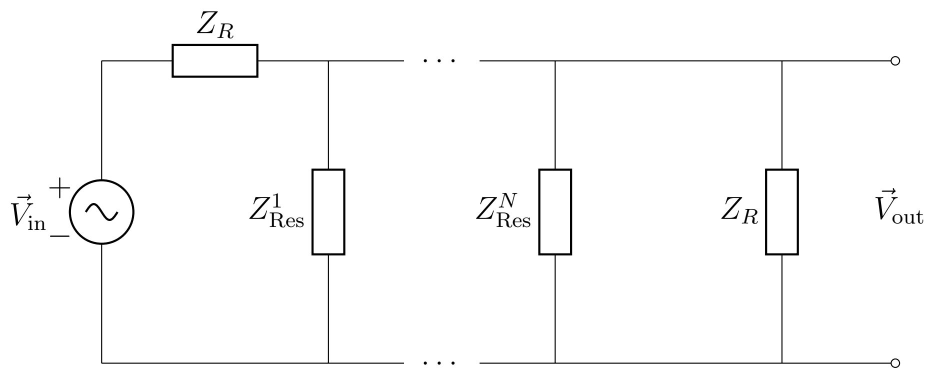


Figure A.3: The circuit shown as a network of impedances, with the resonators drawn as a lumped impedance Z^i_{Res}.

an expression for the transfer function in terms of the impedances of our circuit network.

$$Z_{\text{tot}} = Z_R + \left(\frac{1}{Z_R} + \frac{1}{Z^1_{\text{Res}}} + \ldots + \frac{1}{Z^N_{\text{Res}}}\right)^{-1}$$

$$Z_{\text{tot}} = Z_R + \frac{1}{\frac{1}{Z_R} + \sum_{i=1}^{N} \frac{1}{Z^i_{\text{Res}}}}$$

$$Z_{\text{tot}} = \frac{2 + Z_R \sum_{i=1}^{N} \frac{1}{Z^i_{\text{Res}}}}{\frac{1}{Z_R} + \sum_{i=1}^{N} \frac{1}{Z^i_{\text{Res}}}}$$

$$\frac{\vec{V}_{\text{out}}}{\vec{V}_{\text{in}}} = \frac{\frac{1}{\frac{1}{Z_R} + \sum_{i=1}^{N} \frac{1}{Z^i_{\text{Res}}}}}{\frac{2 + Z_R \sum_{i=1}^{N} \frac{1}{Z^i_{\text{Res}}}}{\frac{1}{Z_R} + \sum_{i=1}^{N} \frac{1}{Z^i_{\text{Res}}}}}$$

$$\vec{H} = \frac{1}{2 + Z_R \sum_{i=1}^{N} \frac{1}{Z^i_{\text{Res}}}}$$

Now that we have an expression for the transfer function, we can work on simplifying it algebraically so that we can gain intuition about the circuit and obtain expressions for the relative phase and amplitude of our transmission wave. Plugging the impedances into our expression for the transfer function, we find

$$\vec{H} = \frac{1}{2 + j\displaystyle\sum_{i=1}^{N} \frac{R\omega(C_t^i + C[V_i(t)])}{1 - \omega^2 L_i C[V_i(t)] - \omega^2 L_i C_t^i}}$$

Multiplying both the numerator and the denominator by the complex conjugate of the denominator, $\vec{H}$ becomes

$$\vec{H} = \frac{2 - j\displaystyle\sum_{i=1}^{N} \frac{R\omega(C_t^i + C[V_i(t)])}{1 - \omega^2 L_i C[V_i(t)] - \omega^2 L_i C_t^i}}{(2)^2 + \left(\displaystyle\sum_{i=1}^{N} \frac{R\omega(C_t^i + C[V_i(t)])}{1 - \omega^2 L_i C[V_i(t)] - \omega^2 L_i C_t^i}\right)^2}$$

$$\vec{H} = \frac{2}{4 + \left(\displaystyle\sum_{i=1}^{N} \frac{R\omega(C_t^i + C[V_i(t)])}{1 - \omega^2 L_i C[V_i(t)] - \omega^2 L_i C_t^i}\right)^2} - j\frac{\displaystyle\sum_{i=1}^{N} \frac{R\omega(C_t^i + C[V_i(t)])}{1 - \omega^2 L_i C[V_i(t)] - \omega^2 L_i C_t^i}}{4 + \left(\displaystyle\sum_{i=1}^{N} \frac{R\omega(C_t^i + C[V_i(t)])}{1 - \omega^2 L_i C[V_i(t)] - \omega^2 L_i C_t^i}\right)^2}$$

Recalling that in complex space

$$|\vec{H}| = \sqrt{\mathrm{Re}(\vec{H})^2 + \mathrm{Im}(\vec{H})^2}$$
$$\angle\vec{H} = \arctan\left(\frac{\mathrm{Im}(\vec{H})}{\mathrm{Re}(\vec{H})}\right)$$

We can use our earlier expressions for A_{out} and ϕ_{out} to obtain an expression for the relative amplitude and phase of the output phasor.

$$|\vec{H}| = \frac{1}{\sqrt{4 + \left(\displaystyle\sum_{i=1}^{N} \frac{R\omega(C_t^i + C[V_i(t)])}{1 - \omega^2 L_i C[V_i(t)] - \omega^2 L_i C_t^i}\right)^2}} \tag{A.1}$$

$$\angle\vec{H} = \arctan\left(\frac{-\displaystyle\sum_{i=1}^{N} \frac{R\omega(C_t^i + C[V_i(t)])}{1 - \omega^2 L_i C[V_i(t)] - \omega^2 L_i C_t^i}}{2}\right) \tag{A.2}$$

A.4 Frequency Response

Armed with our expression for the transfer function of the input phasor to the output phasor, we can characterize the amplitude and phase behavior of a resonator around the frequency it resonates at. Note that these functions are functions of two variables—the frequency ω of the sinusoid on the transmission line, and the voltage $V_i(t)$ applied across the varactor diode. During operation of the sensors, the frequency of the sinusoid on the transmission line is constant, and each resonator is being driven at it's own unique resonant frequency.

However, to get intuition for resonator frequency domain bandwidth and response, it is useful to instead analyze the response of the circuit over a frequency sweep while the voltage across the varactor is constant. For simplicity, we focus on analyzing the frequency response of the i^{th} resonator, and so long as we confine our analysis to the frequency band containing this particular resonator we can safely ignore the effects of other resonators and drop the summations from our equation. In that case we excite the transmission line with a single frequency ω which sweeps the frequency band at an instant in time t_0 such that $V_i(t) = 0$, and thus $C[V_i(t) = 0] = C_0$. Therefore, we can rewrite our expressions for amplitude and phase from the previous section as follows:

$$|\vec{H}(\omega, V_i(t_0) = 0)| = \frac{1}{\sqrt{4 + \left(\frac{R\omega(C_t^i + C_0)}{1 - \omega^2 L_i C_0 - \omega^2 L_i C_t^i}\right)^2}}$$
$$\angle\vec{H}(\omega, V_i(t_0) = 0) = \arctan\left(-\frac{R\omega(C_t^i + C_0)}{2(1 - \omega^2 L_i C_0 - \omega^2 L_i C_t^i)}\right)$$

Though the next section will focus on how to select the coupling capacitance and resonator inductance given the transmission line resistance, varactor diode voltage-capacitance response, and resonance frequency, for now we will just assume that these values are given and show the results of our analytical expressions. We'll also assume that the input phasor has unit amplitude and zero phase to facilitate visualization of the order of the effect of the resonant circuit on these parameters.

A.4.1 Amplitude Response

The amplitude response of the circuit is straightforward. With a varactor diode whose zero bias-voltage capacitance C_0 is 82 pF, a transmission line of resistance R of 2 kΩ, a tuning capacitance C_t of $\sim$ 49 pF, and a $\sim$ 6.5 μH inductance L_i, we can form a resonator with the following frequency response.

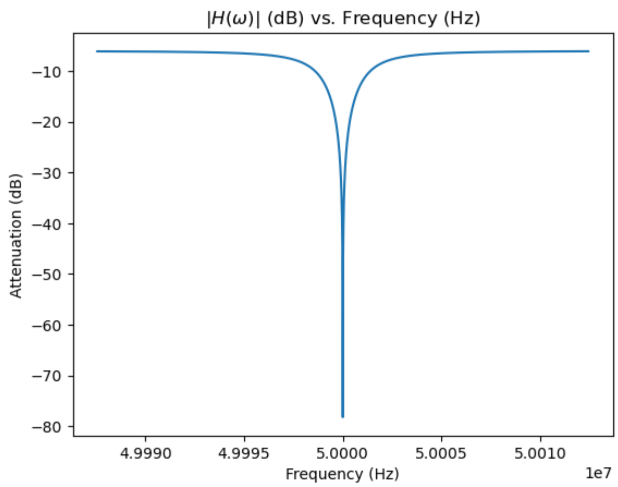


Figure A.4: The relative amplitude of the output phasor as compared to an input phasor with zero phase and unit amplitude applied to the transmission line of a resonator with the following parameters: zero bias-voltage capacitance C_0 of 82 pF, a transmission line of resistance R of 2 kΩ, a coupling capacitance C_c^i of $\sim$49 pF, and a $\sim$ 6.5 μH inductance L_i.

These particular resonator component values have been selected to ensure that the resonant frequency of the resonator is at 172 kHz. The relative amplitude of the output phasor as compared to the input phasor drops sharply around the resonant frequency of the resonator, but is mostly unaffected at frequencies away from that resonant frequency. This is a desired property which enables frequency domain multiplexing—so long as each resonator has it's own band of frequencies which are not encroached upon by other resonators, it won't interfere with other resonators on the same transmission line.

A.4.2 Phase Response

For the sake of simplicity, we'll use the same resonator parameters (zero bias-voltage capacitance C_0 of 82 pF, a transmission line of resistance R of 2 kΩ, a coupling capacitance C_c^i of ∼49 pF, and a $\sim 6.5\ \mu$H inductance L_i) to plot the phase frequency response. However, it's a little bit more complicated to plot the relative phase of the output phasor as compared to the input phasor because the expression involves an arctangent of the imaginary and real parts of the transfer function. However, the period of the tangent function is only from $-\pi/2$ to $\pi/2$, which causes a couple problems when we take the arctangent of phase. The first and most significant problem is that phase is periodic—when we add any multiple of 2π to it, we should get the same phase. The second problem is that with our current analytical expression for the phase, it happens that there is a discontinuity in phase as the frequency sweep passes over the resonant frequency.

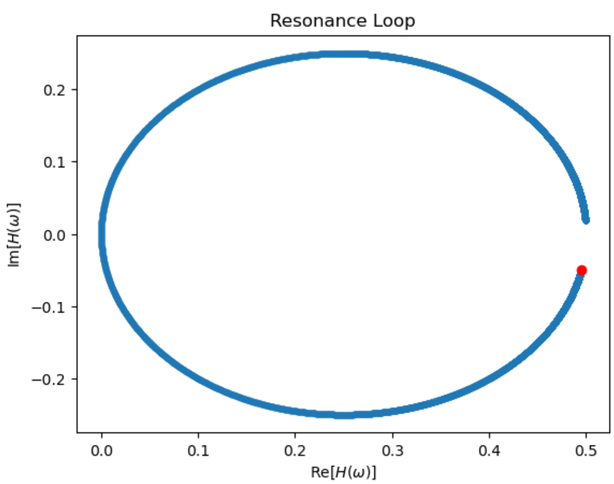


Figure A.5: The transfer function $\vec{H}$ of a resonator with the following parameters: zero bias-voltage capacitance C_0 of 82 pF, a transmission line of resistance R of 2 kΩ, a coupling capacitance C_c^i of ∼49 pF, and a $\sim 6.5\ \mu$H inductance L_i. The red dot indicates the location in the complex plane at the lowest frequency in the frequency sweep.

The first problem can be resolved by redefining our expression for phase to ensure that it has a range of 2π, and the second problem can be resolved by phase wrapping—modolo division of each point by 2π and keeping the remainder will eliminate any discontinuities that arise from crossing over the resonant frequency. To motivate a solution for how to redefine our expression of phase, we plot the transfer function $\vec{H}$ in the complex plane.

From this figure, we can see that a much more natural definition of phase would be to identify the center of the loop that the transfer function sweeps out during a frequency sweep. Then, we can re-define phase $\angle\vec{H} \rightarrow \angle\vec{H}'$ as the angle swept out by a phasor centered at this loop (which notably, is not centered at the origin because the real part of the transfer function is positive definite). Finally, we define zero phase as the angle made by this new phasor when it points to the location of the transfer function in the complex plane corresponding to the resonant frequency of the resonator. This gives our relative phase the expected range of 2π while enabling both negative and positive phase shifts of up to 180 degrees.

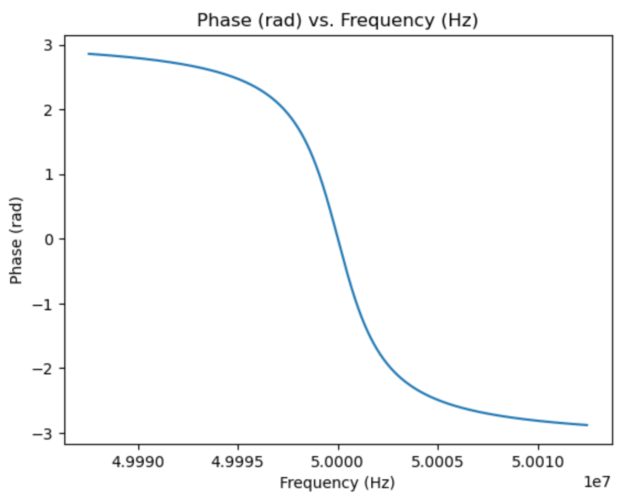


Figure A.6: Relabeled phase $\angle\vec{H}'$ of the output phasor with respect to the input phasor, from a resonator with the following parameters: zero bias-voltage capacitance C_0 of 82 pF, a transmission line of resistance R of 2 kΩ, a coupling capacitance C_c^i of $\sim$49 pF, and a $\sim$ 6.5 μH inductance L_i.

A.5 Resonator Sensitivity & Bandwidth

Now, with the intuition we've gained for how a resonator responds to a frequency sweep, we can focus on analytically determining how to select the tuning capacitance C_t^i and resonator inductance L_i given the transmission line resistance, varactor diode voltage-capacitance response, and desired resonant frequency. Since different varactor diodes have different voltage-capacitance responses, we'll need to work backwards to determine the selectivity of the resonator.

Suppose we have a varactor diode whose capacitance with zero bias voltage is C_0. Then, as the diode is biased, this capacitance will change to some new value, $C_0 + \Delta C(V)$. In other words, suppose the signal we'd like to measure has a maximum amplitude of $V_{\max}$. When this voltage is applied across the varactor, it will result in a corresponding maximum change in capacitance $\Delta C_{\max}$. Note that most varactor diode datasheets' voltage-capacitance curves do not have fine enough resolution to show capacitances resulting from changes on the order of millivolts, which means it will be necessary to characterize the varactor diode's voltage-capacitance response.

To ensure there is sufficient sensitivity in phase to capture these changes in capacitance (and thus, capture these changes in voltage), we need to ensure that our shifts in phase fall within the linear region of the our phase response from Figure A.6. We could accomplish this by working with an expression of the phase, but since that involves an arctangent it gets algebraically messy pretty quickly. Also, there's not a clearly defined cutoff between the linear and asymptotic regions of the phase response, so preferably we'd like a meaningful way to select that arbitrary cutoff point. To do this, we return to equation 2.1 and our expression for the relative amplitude of the output phasor. By picking resonator components so that the full-width half-maximum (FWHM) of the resonance feature corresponds with the maximum change in capacitance, we can be confident that the phase response will fall within the linear region.

To be able to solve for the tuning capacitance and inductance, we'll need a system of two equations. Our first equation will be an expression for the resonant frequency of a resonator, and the second equation will come from our desired resonator sensitivity. To get the first equation, we'll need to write down equation 2.1 for a single resonator, recalling that we excite the transmission line with a sinusoid of constant frequency ω_0^i corresponding to resonance at zero varactor bias voltage. For notational ease, we will drop all the i subscripts and endeavor to remember that every value except for C_0 and R are unique to the resonator in the analysis. Technically, C_0 is also unique to the resonator, but ostensibly with varactors made on the same substrate, process variations will be minimal.

$$|\vec{H}(\omega = \omega_0, V(t_0) = 0)| = \frac{1}{\sqrt{4 + \left(\frac{R\omega_0(C_t + C_0)}{1 - \omega_0^2 LC_0 - \omega_0^2 LC_t}\right)^2}}$$

Recalling our earlier impedance analysis, note that the output phasor amplitude is $\propto \frac{1}{Z^i_{\text{res}}}$. Again, the physical interpretation of this type of resonator at resonant frequency is that the impedance of the resonator drops to zero, causing the entire resonator to act like a short between the transmission line and the ground for signals at that frequency. Shorting the transmission to ground effectively makes the amplitude of the output phasor equal to zero for signals at that frequency. Thus, to get our first equation relating the resonant frequency to the amplitude of the output phasor, we set the above expression equal to zero. Since the numerator is positive definite, the only possible way for $\vec{H}$ to become zero is in the limit in which the denominator approaches infinity. In order for the denominator of $\vec{H}$ to be equal to infinity, then

$$\begin{aligned} 0 &= 1 - \omega_0^2 LC_0 - \omega_0^2 LC_t \\ \omega_0^2 &= \frac{1}{L(C_0 + C_t)} \end{aligned}$$

As expected, we recover the expression for the resonant frequency of a standard tank resonator, $\omega_0^2 = \frac{1}{LC}$, with the capacitance of the varactor diode and the tuning capacitor added together. Now that we have an expression for the resonant frequency, we can focus on obtaining an expression for resonator sensitivity to a change in capacitance from C_0 by $\pm\Delta C_{\max}$ using equation 2.1 when $V(t) = \pm V_{\max}$, where this is the maximum voltage swing the sensor is expected to see. The maximum possible value of $\vec{H}$ is 1/2 (which intuitively corresponds to the case in which all the resonators are far from resonance and have functionally infinite impedance, and thus all the current from the input goes through two resistors of equal value, creating a voltage divider) so then the FWHM of the resonance feature corresponds with when $\vec{H}$ is 1/4.

$$
\begin{aligned}
|\vec{H}(\omega = \omega_0, V(t_0) = \pm V_{\max})| &= \frac{1}{\sqrt{4 + \left(\frac{R\omega_0(C_t + C_0 \pm \Delta C_{\max})}{1 - \omega_0^2 L[C_0 + \Delta C_{\max}] - \omega_0^2 L C_t}\right)^2}} \\
\frac{1}{4} &= \frac{1}{\sqrt{4 + \left(\frac{R\omega_0(C_t + C_0 \pm \Delta C_{\max})}{1 - \omega_0^2 L[C_0 + \Delta C_{\max}] - \omega_0^2 L C_t}\right)^2}} \\
16 &= 4 + \left(\frac{R\omega_0(C_t + C_0 \pm \Delta C_{\max})}{1 - \omega_0^2 L[C_0 + \Delta C_{\max}] - \omega_0^2 L C_t}\right)^2 \\
\sqrt{12} &= \frac{R\omega_0(C_t + C_0 \pm \Delta C_{\max})}{1 - \omega_0^2 L[C_0 + \Delta C_{\max}] - \omega_0^2 L C_t}
\end{aligned}
$$

We've now written down two equations for our resonators in terms of constants (R, C_0), parameters we determine during the design process of a resonator (desired resonant frequency ω_0 and maximum change in capacitance $\Delta C_{\max}$), and the resonator specific values we'd like to solve for (L, C_t). By re-writing our first equation for the resonant frequency to get L in terms of C_c, we can substitute this into our second equation and solve for C_t.

$$
\begin{aligned}
L &= \frac{1}{\omega_0^2(C_0+C_t)} \\
\sqrt{12} &= \frac{R\omega_0(C_t+C_0\pm\Delta C_{\max})}{1-\omega_0^2L[C_0\pm\Delta C_{\max}]-\omega_0^2LC_t} \\
\sqrt{12} &= \frac{R\omega_0(C_t+C_0\pm\Delta C_{\max})}{1-\frac{C_0\pm\Delta C_{\max}+C_t}{C_0+C_t}} \\
\sqrt{12} &= \frac{R\omega_0(C_t+C_0\pm\Delta C_{\max})(C_0+C_t)}{C_0+C_t-(C_0\pm\Delta C_{\max}+C_t)} \\
\sqrt{12} &= \mp\frac{R\omega_0}{\Delta C_{\max}}(C_t^2+C_0^2+2C_tC_0\pm\Delta C_{\max}C_0\pm\Delta C_{\max}C_t) \\
0 &= \mp\frac{R\omega_0}{\Delta C_{\max}}C_t^2\mp\frac{R\omega_0}{\Delta C_{\max}}(2C_0\pm\Delta C_{\max})C_t-\sqrt{12}\mp\frac{R\omega_0}{\Delta C_{\max}}(C_0^2\pm\Delta C_{\max}C_0) \\
0 &= C_t^2+(2C_0\pm\Delta C_{\max})C_t\pm\frac{\Delta C_{\max}\sqrt{12}}{R\omega_0}+C_0^2\pm\Delta C_{\max}C_0
\end{aligned}
$$

With no loss of generality (ie: by appealing to symmetry about the resonant frequency), we can select the negative sign of $\Delta C_{\max}$ corresponding to $-V_{\max}$, since selecting the positive sign leads to an imaginary solution at low resonant frequencies. Solving this quadratic for C_t and plugging it into our expression for the inductance gives us the following:

$$
C_t = \frac{\Delta C_{\max}}{2} - C_0 + \sqrt{\frac{\Delta C_{\max}^2}{4}+\frac{\sqrt{12}\Delta C_{\max}}{R\omega_0}} \tag{A.3}
$$

$$
L = \frac{1}{\omega_0^2(C_t+C_0)} \tag{A.4}
$$

Notably, a lower limit is placed on the sensitivity of the resonator because the tuning capacitance C_t is only positive when

$$
\Delta C_{\max} > \frac{C_0^2R\omega_0}{C_0R\omega_0+\sqrt{12}}
$$

To calculate how much bandwidth the resonator is going to consume under these design specifications, we can plug in our calculated values for C_t and L into our expression for $\vec{H}$, then solve for the frequency at which the FWHM occurs while no voltage is applied across the varactor diode. Since we already know the resonant frequency, the solution will give us a conservative estimate to how much frequency bandwidth the resonator will consume.

$$
\begin{aligned}
|\vec{H}(\omega, V(t_0) = 0)| &= \frac{1}{\sqrt{4 + \left(\frac{R\omega(C_t + C_0)}{1 - \omega^2 LC_0 - \omega^2 LC_t}\right)^2}} \\
\frac{1}{4} &= \frac{1}{\sqrt{4 + \left(\frac{R\omega(C_t + C_0)}{1 - \omega^2 LC_0 - \omega^2 LC_t}\right)^2}} \\
16 &= 4 + \left(\frac{R\omega(C_t + C_0)}{1 - \omega^2 LC_0 - \omega^2 LC_t}\right)^2 \\
\sqrt{12} &= \frac{R\omega(C_t + C_0)}{1 - \omega^2 LC_0 - \omega^2 LC_t} \\
0 &= L\sqrt{12}(C_0 + C_t)\omega^2 + R(C_0 + C_t)\omega - \sqrt{12} \\
0 &= L\sqrt{12}\omega^2 + R\omega - \frac{\sqrt{12}}{C_0 + C_t}
\end{aligned}
$$

Solving this quadratic for ω will yield the two frequencies (one on either side of the resonant frequency) for which the amplitude drops to half the maximum. The roots are given by

$$
\omega_{\pm} = \frac{-R \pm \sqrt{R^2 + \frac{48L}{C_0 + C_t}}}{2L\sqrt{12}}
$$

Because technically we should get four roots (due to the square root in our original expression for $\vec{H}$), we end up needing to flip the sign of the ω_- root to get two positive values to subtract from one another. Thus we get the following expression

$$
\begin{aligned}
\Delta\omega_{\text{FWHM}} &= \omega_- - \omega_+ \\
\Delta\omega_{\text{FWHM}} &= \frac{R}{L\sqrt{12}}
\end{aligned}
$$

We can rewrite L in terms of C_t and ω_0 using equation 2.4 to understand the contribution of that component/setpoint to the bandwidth consumed.